World Layoffs 2020–2023

Exploratory Data Analysis Report

Database world_layoff	**Table** layoff_staging1 (cleaned)
Raw Rows 2,361	**Clean Rows** 1,995 (366 rows removed — 15.5%)
Scope Global — All Sectors	**Period** March 2020 — March 2023

1. Dataset Overview

This report presents a comprehensive exploratory data analysis of global workforce layoffs between March 2020 and March 2023. The dataset captures 1,995 layoff events across 51 countries, 31 industries, and 1,628 companies following the data cleaning process.

1.1 Data Schema

Column	Type	Description
company	VARCHAR	Name of the company that conducted layoffs
location	VARCHAR	City where the company is headquartered
country	VARCHAR	Country of the company
industry	VARCHAR	Business sector / industry vertical
stage	VARCHAR	Funding stage (Seed, Series A–J, Post-IPO, Acquired)
date	DATE	Date the layoff event was reported
total_laid_off	INT	Number of employees laid off in the event
percentage_laid_off	FLOAT	Proportion of workforce laid off (0.0–1.0)
funds_raised_millions	FLOAT	Total funding raised by the company (USD millions)

1.2 Data Quality

The null checker revealed three columns with significant missing values that affect analysis coverage:

Column	Total Rows	Null Count	Null %
percentage_laid_off	1,995	423	21.2%
total_laid_off	1,995	378	18.9%
funds_raised_millions	1,995	165	8.3%
date	1,995	1	0.1%
company / country / industry / location / stage	1,995	0	0.0%

> *Rows where both total_laid_off and percentage_laid_off are NULL were removed during cleaning, accounting for the 366 deleted rows (15.5% of raw data). Analysis of total_laid_off uses 1,617 valid rows; percentage_laid_off uses 1,572 valid rows.*

2. Key Performance Indicators

The following KPIs provide a high-level summary of the dataset. All metrics exclude NULL values in their respective columns.

Measure	Value
Number of Companies	1,628
Number of Industries	31
Number of Countries	51
Total Layoffs	383,659
Average Layoffs	237
Total Funds Raised	$1.6 Trillion
Average Funds Raised	$875.1 M
Average Layoff Rate	25.8%

2.1 Workforce Estimation

Using the relationship: Estimated Employees = total_laid_off / percentage_laid_off, an approximate total workforce was computed for companies that reported both fields.

Estimated Total Employees	Total Laid Off (subset)	Overall Layoff Rate
~4,017,000	**325,399**	**8.1%**

> *The 8.1% overall layoff rate means approximately 1 in 12 employees in companies covered by this dataset lost their job during this period. The average event-level layoff rate of 25.8% indicates that when a company did lay off, it was a substantial reduction.*

3. Time-Series Analysis

3.1 Annual Summary

2022 recorded the highest total layoffs (160,661), driven by 1,030 events. 2023 shows a concentration effect: fewer events (397) but higher individual event sizes — the largest single event (12,000 at Google) occurred in 2023.

Year	Total Laid Off	Events	Largest Event	YoY Growth	Annual Share
2022	160,661	1,030	11,000	+915.4%	44.8%
2023	125,677	397	12,000	-21.8%	29.4%
2020	80,998	532	4,375	—	21.1%
2021	15,823	35	2,434	-80.5%	4.7%

> *2021 was an anomalous calm year with only 35 events and 15,823 total layoffs — a -80.5% drop from 2020. This reflects the post-pandemic hiring boom when tech companies aggressively expanded headcount, temporarily reversing layoff trends.*

3.2 Month-over-Month Analysis (selected months)

The table below highlights the most significant monthly inflection points. Full MoM data covers 36 months from March 2020 to March 2023.

Month	Companies	Total Laid Off	MoM Change	MoM Growth %
2020-03	93	9,628	—	—
2020-04	202	26,710	+17,082	+177.4%
2020-05	85	25,804	-906	-3.4%
2020-06	41	7,627	-18,177	-70.4%
2020-07	20	7,112	-515	-6.8%
2022-10	72	17,406	+11,525	+196.0%
2022-11	133	53,451	+36,045	+207.1%
2022-12	81	10,329	-43,122	-80.7%
2023-01	166	84,714	+74,385	+720.2%
2023-02	112	36,493	-48,221	-56.9%
2023-03	21	4,470	-32,023	-87.8%

> *January 2023 stands out as the single most impactful month: 84,714 layoffs across 166 companies, representing a +720.2% jump from December 2022. This was driven by simultaneous announcements from Google (12,000), Microsoft (10,000), Amazon (8,000), and Salesforce (8,000).*

3.3 Cumulative Rolling Total

The rolling total reveals three distinct phases: a sharp COVID-driven rise in mid-2020 (0 to ~81K), a prolonged plateau through 2021 (~81K to ~97K), and an explosive acceleration from late 2022 through early 2023 (~97K to 383K).

Month	Monthly Total	Cumulative Total

2020-03	9,628	9,628
2020-12	852	80,998
2021-12	1,200	96,821
2022-06	17,394	141,137
2022-11	53,451	247,153
2022-12	10,329	257,482
2023-01	84,714	342,196
2023-02	36,493	378,689
2023-03	4,470	383,159

Milestone The dataset crossed 100,000 cumulative layoffs in February 2022, 200,000 in October 2022, and reached 383,159 by March 2023 — meaning 75% of all layoffs in this dataset occurred in just the final 18 months.

3.4 Annual Share of Total Layoffs

Year	Total Laid Off	Avg/Event	Total Funds Raised	Annual Share
2022	150,349	197	$938.4B	44.8%
2023	98,445	381	$222.8B	29.4%
2020	70,755	161	$278.6B	21.1%
2021	15,810	527	$37.8B	4.7%

> *Note: Annual Share figures here differ slightly from Section 3.1 because this query filters on funds_raised_millions IS NOT NULL, reducing the row count. These figures reflect only companies that reported funding data.*

4. Industry Analysis

4.1 Most Affected Industries (Top 10)

Consumer and Retail sectors collectively account for over 88,000 layoffs, reflecting the post-pandemic correction in e-commerce and digital consumer platforms. Finance had the highest number of affected companies (169) but a relatively low average per event (141), suggesting widespread but smaller-scale cuts.

Industry	Companies	Total Laid Off	Avg Per Event
Consumer	72	45,182	525
Retail	103	43,613	330
Other	70	36,289	459
Transportation	80	33,748	309
Finance	169	28,344	141
Healthcare	105	25,953	216
Food	74	22,855	243
Real Estate	51	17,565	231
Travel	36	17,159	357
Hardware	9	13,828	1,382

> *Hardware has the highest average layoff per event (1,382) despite only 9 companies. This is driven by a small number of very large hardware manufacturers making single large-scale reductions rather than multiple smaller rounds.*

4.2 Funding Stage Analysis

Post-IPO companies dominate total layoffs with 173,034 — nearly 10x more than the next stage (Series C at 20,017). This reflects that publicly listed companies are most exposed to investor pressure, interest rate sensitivity, and stock price corrections.

Stage	Companies	Total Laid Off	Avg/Event	Funds Raised
Post-IPO	199	173,034	624	$945.9B
Series C	178	20,017	99	$39.6B
Series D	140	18,985	114	$61.0B
Acquired	84	18,219	195	$29.1B
Series B	193	15,095	72	$20.1B
Series E	77	12,697	138	$51.7B
Series F	36	9,915	206	$41.7B

> *The average layoff per event for Post-IPO companies (624 employees) is the highest among all stages, reinforcing that public market pressures drive larger, more decisive workforce reductions compared to early-stage startups.*

4.3 Cohort Analysis — Stage by Year

In all four years, Post-IPO companies were the largest source of layoffs. Early-stage (Series A/B/Seed) companies become relatively more prominent in 2022 as venture funding dried up and startups faced cash constraints.

Year	Stage	Laid Off	Note
2020	Post-IPO	22,672	Largest — COVID shock hits public cos.
2020	Acquired	12,169	M&A targets also affected
2021	Acquired	2,984	Quiet year; M&A-driven consolidation
2021	Post-IPO	2,895	Minimal — hiring boom in effect
2022	Post-IPO	79,373	Dominant — rate hikes hit valuations
2022	Series C	13,072	VC funding tightens at growth stage
2023	Post-IPO	98,692	Peak — Big Tech mass layoffs
2023	Acquired	6,966	Second largest — restructuring post-M&A

5. Company Analysis

5.1 Top 10 Companies by Total Layoffs

Amazon leads with 18,150 layoffs across 3 separate rounds, making it the most prolific single company in this dataset. The top 10 companies collectively account for approximately 96,465 layoffs — roughly 25% of the dataset total.

#	Company	Industry	Country	Laid Off	Rounds
1	**Amazon**	Retail	United States	18,150	3
2	**Google**	Consumer	United States	12,000	1
3	**Meta**	Consumer	United States	11,000	1
4	**Salesforce**	Sales	United States	10,090	4
5	**Microsoft**	Other	United States	10,000	1
6	**Philips**	Healthcare	Netherlands	10,000	2
7	**Ericsson**	Other	Sweden	8,500	1
8	**Uber**	Transportation	United States	6,700	2
9	**Dell**	Hardware	United States	6,650	1
10	**Booking.com**	Travel	Netherlands	4,375	1

5.2 Top 5 Companies Per Year

The leadership changes significantly across years, reflecting different economic pressures. COVID-era victims (Uber, Airbnb, Booking.com) dominated 2020; the post-pandemic correction brought Big Tech (Meta, Google, Microsoft) to the forefront in 2022–2023.

Year	Rank	Company	Industry	Laid Off
2020	1	**Uber**	Transportation	7,525
2020	2	**Booking.com**	Travel	4,375
2020	3	**Groupon**	Retail	2,800
2020	4	**Swiggy**	Food	2,250
2020	5	**Airbnb**	Travel	1,900
2021	1	**Bytedance**	Consumer	3,600
2021	2	**Katerra**	Construction	2,434
2021	3	**Zillow**	Real Estate	2,000
2021	4	**Instacart**	Food	1,877
2021	5	**WhiteHat Jr**	Education	1,800
2022	1	**Meta**	Consumer	11,000
2022	2	**Amazon**	Retail	10,150
2022	3	**Cisco**	Infrastructure	4,100
2022	4	**Peloton**	Fitness	4,084
2022	5	**Philips**	Healthcare	4,000

2023	1	**Google**	Consumer	12,000
2023	2	**Microsoft**	Other	10,000
2023	3	**Ericsson**	Other	8,500
2023	4	**Amazon**	Retail	8,000
2023	4	**Salesforce**	Sales	8,000

5.3 Complete Shutdowns (100% Laid Off)

Companies with percentage_laid_off = 1.0 represent complete closures — either bankruptcy, acquisition with full workforce elimination, or total business cessation. The table below shows the top 8 by funds raised, illustrating significant capital destruction.

Company	Industry	Country	Funds Raised	Date
Britishvolt	Transportation	United Kingdom	$2.4B	2023-01-17
Quibi	Media	United States	$1.8B	2020-10-21
Deliveroo Australia	Food	Australia	$1.7B	2022-11-15
Katerra	Construction	United States	$1.6B	2021-06-01
BlockFi	Crypto	United States	$1.0B	2022-11-28
Aura Financial	Finance	United States	$584M	2021-01-11
Openpay	Finance	Australia	$299M	2023-02-07
Pollen	Marketing	United Kingdom	$238M	2022-08-10

Key Finding Britishvolt (EV batteries, UK) shut down after raising $2.4B — the largest single capital loss among complete closures. Quibi (streaming) raised $1.8B and shut down within 2 years of launch. These represent failures of high-burn business models in a rising interest rate environment.

5.4 Statistical Outliers

Companies laying off more than 2x the global average (2 x 237 = 474 employees) are classified as outliers. These 148 events account for a disproportionate share of total layoffs and can skew mean-based metrics. Key outliers include: Google (12,000), Meta (11,000), Microsoft (10,000), Amazon (10,000), Ericsson (8,500), and Dell (6,650).

> *The presence of outliers explains why the mean layoff per event (237) differs substantially from the median, which would be significantly lower. Median-based analysis would better represent the 'typical' layoff event.*

6. Geographic Analysis

6.1 Country Rankings

The United States accounts for 66.9% of all layoffs in the dataset (256,559 out of 383,659), reflecting both its dominance in the global tech sector and the concentration of Post-IPO companies subject to market pressure. India is a distant second at 9.4%.

Country	Companies	Total Laid Off	Avg/Event	Global Share
United States	844	256,559	249	66.9%
India	96	35,993	285	9.4%
Netherlands	8	17,220	1,913	4.5%
Sweden	14	11,264	704	2.9%
Brazil	54	10,391	157	2.7%
Germany	46	8,701	158	2.3%
United Kingdom	43	6,398	139	1.7%
Canada	68	6,319	78	1.6%
Singapore	19	5,995	272	1.6%
China	7	5,905	656	1.5%

> *The Netherlands ranks 3rd despite only 8 companies because it is home to large multinationals (Booking.com with 4,375; Philips with 10,000) that each conducted very large single events — hence the highest average per event (1,913).*

6.2 Key City-Level Findings

Within the United States, SF Bay Area dominates with 109,168 layoffs from 248 companies. Seattle follows with 33,936, driven primarily by Amazon and Microsoft headquarters operations. New York City ranks third at 21,727.

Country — City	Companies	Laid Off	Avg/Event	Layoff Rate
US — SF Bay Area	248	109,168	358	9.2%
US — Seattle	28	33,936	893	3.5%
US — New York City	108	21,727	172	15.4%
India — Bengaluru	29	14,039	360	9.7%
India — Gurugram	9	3,776	378	15.3%
Germany — Walldorf	1	3,000	3,000	3.0%
Sweden — Stockholm	6	10,370	1,481	8.1%
Brazil — Sao Paulo	28	5,660	157	10.8%

7. Correlation Analysis

7.1 Pearson Correlation: Funds Raised vs. Layoffs

A Pearson correlation coefficient was computed between total funds raised (funds_raised_millions) and the number of employees laid off (total_laid_off) at the event level.

Result Pearson r = 0.077 → Weak positive correlation

Interpretation of the correlation coefficient:

Range	Interpretation	Implication
0.0 – 0.3	Weak	Funding has little effect on layoff size
0.3 – 0.6	Moderate	Partial relationship exists
0.6 – 1.0	Strong	Higher funding → larger layoffs
0.077 (our result)	**Weak positive**	**No meaningful linear relationship**

> *The near-zero correlation (r = 0.077) indicates that the amount of funding a company raised does NOT predict how many employees it will lay off. A startup that raised $10M and one that raised $10B are equally likely to lay off a similar proportion. This challenges the assumption that 'well-funded companies cut more.' The relationship between funding and layoffs is non-linear and likely mediated by factors such as sector, stage, and macroeconomic timing.*

7.2 Limitations of Correlation Analysis

Several factors limit the interpretability of this Pearson correlation:

- Outliers: A small number of very large companies (Amazon, Google, Meta) with high funding and high layoffs inflate the numerator without creating a true linear trend.
- Non-linearity: The relationship between funding and layoffs may be U-shaped — very small startups and very large corporations both lay off, while mid-stage companies may be more conservative.
- Survivorship bias: Companies that fully shut down may not have reported complete funding data, underrepresenting the most severe cases.
- Temporal confounding: Layoffs from 2022–2023 are driven by macroeconomic conditions (rate hikes), not funding levels per se.

8. Conclusions & Key Takeaways

8.1 Summary of Findings

#	Dimension	Key Finding
1	Volume	383,659 total layoffs across 1,995 events; avg 237 per event
2	Timing	75% of all layoffs occurred in the 18 months from Oct 2022 to Mar 2023
3	Peak Month	January 2023: 84,714 layoffs (+720% MoM) — single largest month
4	Hardest Year	2022: 160,661 total layoffs across 1,030 events (44.8% of dataset)
5	Top Industry	Consumer sector: 45,182 layoffs; Hardware: highest avg per event (1,382)
6	Top Company	Amazon: 18,150 across 3 rounds — most rounds and highest total
7	Geography	US: 66.9% of global layoffs; SF Bay Area: 109,168 — single largest city
8	Stage	Post-IPO companies: 173,034 layoffs — 45% of all layoffs from 1 stage
9	Shutdowns	Britishvolt ($2.4B raised) and Quibi ($1.8B) — largest capital destructions
10	Correlation	Funds raised does NOT predict layoff size (r = 0.077)

8.2 Macroeconomic Context

The data strongly reflects three macroeconomic phases:

1. Phase 1 (Mar–Jul 2020): COVID-19 shock — immediate sector-specific layoffs in travel, food, and consumer services.
2. Phase 2 (Aug 2020–Dec 2021): Recovery and over-hiring — tech companies aggressively expanded to meet digital demand, suppressing layoffs.
3. Phase 3 (Jan 2022–Mar 2023): Correction — Federal Reserve rate hikes (0.25% → 5.5% in 15 months), declining ad revenue, and valuation compression triggered mass layoffs in Post-IPO and late-stage companies.

8.3 Limitations

This analysis is subject to the following limitations: (1) 18.9% of events have no total_laid_off value, excluding them from quantitative analysis; (2) the dataset covers only reported events — unreported or informal layoffs are absent; (3) geographic assignment is based on company HQ, not the location of affected employees; (4) the dataset ends in March 2023, excluding subsequent developments.

Appendix — SQL Queries Reference

All findings in this report were derived from the following SQL queries executed against layoff_staging1 in the world_layoff database on SQL Server. Queries are categorized by analysis section.

4. Section 1 — Row Counts & Data Duration
5. Section 2 — Universal Null Checker (Dynamic SQL)
6. Section 3 — KPI Summary (Part A: Metrics, Part B: Workforce Estimation)
7. Section 4 — Time Analysis (Q1: Annual, Q2: YoY, Q3: MoM, Q4: Rolling Total, Q5: Annual Share)
8. Section 5 — Industry Analysis (By Industry, By Stage, Cohort Analysis)
9. Section 6 — Company Analysis (Top 10, Shutdowns, Outliers, Top 5 Per Year)
10. Section 7 — Geographic Analysis (By Country, By City)
11. Section 8 — Correlation Analysis (Pearson r: Funds Raised vs. Layoffs)